MILITARY SPECIFICATION

INTRAVENOUS INJECTION SET, WITH FLEXIBLE DRIP CHAMBER, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, disposable intravenous injection sets (infusion apparatus) designed for use in the administration of a single parenteral solution from a standard flexible plastic solution container.

1.2 Classification. Intravenous injection sets shall be of the following types as specified (see 6.2):

Type I. - Regular set (15-20 drops/ml.) without Needle

Type II. - Pediatric set (60 drops/ml.) without Needle

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Paper, (for Carton Sealing).

Military

MIL-S-36586	Sterilization strip test, Bacterial Spore.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
Bureau of Biologics. Food and Drug Administration
Code of Federal Regulation Part 273. Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, MD 20852.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI Z70.1	American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain not more than 5 parts per million (p.p.m.) total of lead, barium, and cadmium when tested as specified in 4.4.1, or 4.4.1.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (USP) for Class II plastics and for transfusion and infusion assemblies when tested as specified in 4.4.2.

3.1.1.1 Plastic tubing. Plastic tubing shall be clear, colorless, transparent, flexible polyvinyl chloride. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be clean and free of particles and shall be free from all liquid lubricating substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.2 Other material. All other material hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Type I - Regular set. The type I set shall be a sterile, nonpyrogenic, safe (nontoxic), closed system disposable intravenous injection set (infusion device) for use in the administration of a single parenteral solution from a flexible plastic solution container (see 3.2.2). The style and design shall be in accordance with Figure 1 and as specified herein. The set shall have a cap with piercing device, a flexible plastic calibrated drip chamber, one female needle adapter a length of plastic tubing, an injection site, a male needle adapter, a flow control device, and end protectors. The drip chamber shall be made of suitable, flexible, colorless clear plastic. The base of the cap shall be permanently attached to the drip chamber in a leakproof manner. The assembled drip chamber shall be so shaped as to provide a convenient hand grip for the insertion of the piercing device into the port of the solution container. The assembly of the set shall be structurally sound and shall not leak air or fluid during use. Open ends of the set shall be tightly covered (sealed) with readily removable covers, or end protectors, which shall maintain the sterility and freedom from pyrogens of the fluid path after sterilization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

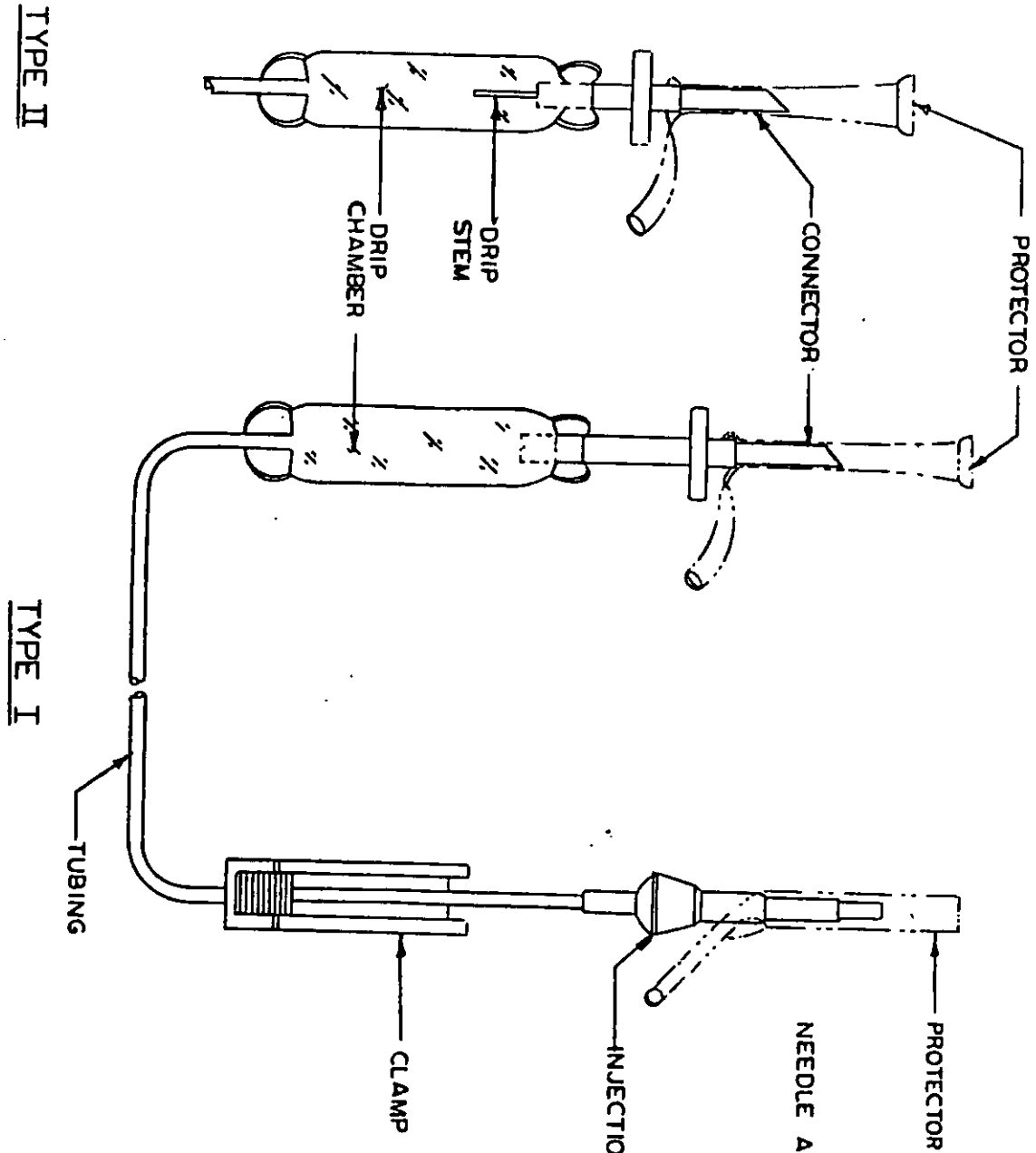


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

3.2.1 Drip chamber. The drip chamber housing shall have a volume between 4 ml minimum and 5.5 ml maximum.

3.2.2 Cap (piercing device). The cap shall consist of an integrally designed unit including a piercing device (spike). The piercing device shall be suitable rigid plastic, spike, with the other channel being extended to form the drip orifice. The piercing device shall be not less than 5/8 inch in length, with a diameter of nominal 1/4 inch at the base (proximal end). It shall be firmly retained in a leakproof manner when inserted in the port of a standard flexible plastic solution container. The piercing device shall not break or leak air or fluid during use. It shall not be brittle or soft, or become brittle or soft under normal storage conditions. The fluid passage in the spike shall extend into the drip chamber, and shall be calibrated to deliver 15-20 drops per milliliter (ml.) when tested as specified in 4.4.3. Shall maintain the 15-20 drops per milliliter (ml.) when tested as specified in 4.4.3. Shall maintain the 15-20 drops per ml drop size during administration of intravenous solutions for a minimum of 48 hours at approximately 60 drops per minute.

3.2.2.1 Durability of piercing device. When tested as specified in 4.4.4 the device, or the cap, shall not loosen or permit leakage, nor the device or cap become damaged. The piercing device shall be capable of being inserted into the standard solution containers, as described in 4.4.4, without difficulty, the tip (or point) of the device not breaking or bending to the degree of causing difficulty in insertion into standard solution containers.

3.2.3 Plastic tubing. The plastic tubing shall be fabricated of material as specified in 3.1.1.1. It shall be firmly and permanently secured, in a leakproof manner, to the drip chamber at one end and to a male needle adapter at the other end in such a manner as to form an airtight, leakproof connection which shall not separate when tested as specified in 4.4.8. The overall length of the set shall be a minimum of 71 inches.

3.2.4 Injection site. The injection site shall be ball shaped and located adjacent to the needle adapter, molded from natural rubber with targeted, resealable, needle puncture sites.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

3.2.5 Control clamp. Control clamp shall be fabricated of plastic, a screw-type or roller clamp from which the screw or roller cannot be accidentally removed. The clamp shall be supplied on tubing in open position and shall provide a constant setting and quick shut off from the operating position, shall be movable, and shall be capable of one handed operation. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed drastically by an accidental jerk on the tubing. When tested as specified in 4.4.7, the clamp shall be capable of accomplishing complete quick shutoff (i.e. no falling drop in drip chamber or from needle adapter (3.2.6), shall regulate flow of the solution from complete shutoff to maximum flow, and with a constant head of water shall maintain a constant flow rate at any point between complete shutoff and maximum flow. In addition, when the clamp is operated from its fully open position to complete shutoff, and this operation is repeated ten times while the clamp is in the same position on the tubing, the tubing shall not be cut, pierced, or otherwise damaged so as to cause leakage or restriction of flow of solution.

3.2.6 Male needle adapter. The male needle adapter shall be fabricated of suitable translucent plastic and shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI Standard Z70.1 in a tight leakproof manner (see 3.8). The needle adapter shall not separate from the molded rubber part or show evidence of leakage when tested as specified in 4.4.5.

3.2.7 End protector. The protector shall be attached to the needle adapter while the needle is contained within the protector in a manner which will prevent contact of the needle point with the inside end of the protector and also provide a loose fit over the Luer end of the needle adapter. End protector shall be readily removable, transparent or semi-transparent plastic tube, or sheath constructed in such a manner that when placed over each open end of the intravenous injection set it shall, after sterilization, maintain sterility, freedom from pyrogens, and the integrity of the entire fluid path of the infusion device until manually removed.

3.3 Type II - Pediatric set. Type II pediatric intravenous injection set shall comply with the requirements specified herein for the type I (Regular) set (see 3.2) except shall contain a plastic calibrated drip chamber with a metal drip stem cannula of such a size and shape and shall be calibrated to deliver 60 ± 9 drops per milliliter when tested as specified in 4.4.3. Shall maintain the 60 drop per milliliter drop size during administration of intravenous solutions for a minimum of 48 hours at approximately 60 drops per minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

3.4 Manufacturers' record. The manufacturer shall place a lot or control number on the paper label, or on the printed carbon, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, U.S. Department of Health, Education and Welfare, any official of the Federal Food and Drug Administration, or any duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent record of the manufacturer.

3.5 Sterilization, safety, and freedom from pyrogens. All equipment shall be safe. Fluid paths, and end connectors of type I sets and fluid paths, and end connectors of type II sets shall be sterile and pyrogen-free. All the above shall be in accordance with current revisions of the minimum requirements of the National Institutes of Health for citrated whole blood. The fluid path and end connectors shall be pyrogen-free when tested as specified in 4.4.8 and sterile when tested with a biological indicator as specified in 4.4.9. The sets shall be safe when tested for safety as specified in 4.4.10.

3.6 Certificates. Certificates of sterility, freedom from pyrogens, safety, and bacterial retention as specified below, shall be submitted. For destination inspection, certificates shall be submitted by the contractor directly to Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, Attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection certificates shall be furnished to the procuring activity via the government inspector.

3.6.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organsim(s) and population density used.
j. Test medium, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

3.6.1.1 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.6.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.5 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.6.3 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.5, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.7 Compatibility. The materials of which the type I and type II sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there by any materials incorporated in the set which will cause any deleterious effect when the set is used for the purpose intended.

3.8 Leakage and separation of set. Type I and type II sets shall be tested for leakage and separation of set as specified in 4.4.5 and 4.4.5.1. Sets which show leakage or separation at any part of the set shall be rejected. A leak is defined as the formation of an air (gas) bubble or liquid as appropriate, which breaks away from the set within the specified test time (5 seconds).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

3.9 Instructions. Individual set carton shall have detailed instructions for proper use of the set for single use. Detailed instruction sheet for tandem use shall be inserted in each unit package (48's).

3.10 Workmanship. Workmanship shall be first class throughout. The intravenous injection sets shall be free from defects which detract from their appearance or impair their serviceability.

3.11 Government furnished material. The Government shall furnish the contractor the dextrose injection bags (National Stock No. (NSN) 6505-00-083-6538) needed to perform the leakage and drip test specified in 4.4.5.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the components or materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

4.2 Sampling.

4.2.1 Lot.

4.2.1.1 Plastic. For purposes of this specification, a lot or batch shall consist of a quantity of plastic moldings or tubing of uniform composition produced under like conditions.

4.2.2 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one intravenous injection set.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.3 For tests.

4.2.3.1 For sterility test. Sampling for sterility test shall be in accordance with the United States Pharmacopeia (USP).

4.2.3.2 For pyrogen, and safety tests. Sampling procedures for pyrogen and safety tests shall be conducted as indicated in the USP, for transfusion and infusion assemblies.

4.2.3.3 For extractable metals test. Samples shall be taken from each lot of rubber or plastic as indicated in 4.4.

4.2.4 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

Table II. Sampling for Other Tests

Component (unit of product)	Characteristic	Require-ment	Test procedure	Insp. level	AQL (percent defective)
Plastic and rubber components	Heavy metal (extractable)	3.1.1	4.4.1	**	*
	Biological	3.1.1	4.4.2	**	*
1-Intravenous injection set	Dimensions	3.2	-----	S-3	*
1-Intravenous injection set	Drop test	3.2.2 and 3.3	4.4.3	S-3	*

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

Table II. Sampling for Other Tests (cont'd)

Component (unit of product)	Characteristic	Requirement	Test procedure	Insp. level	AQL (percent defective)
1-Intravenous injection set	Piercing device durability	3.2.2.1	4.4.4	S-3	*
1-Intravenous injection set	Leakage, and separation of set	3.2.2, 3.2.6 and 3.8	4.4.5 and 4.4.5.1	I, except sample size shall be a minimum of 125 samples	*
1-Intravenous injection set	Reseal (rubber diaphragm)	3.2.4	4.4.6	S-3	*
1-Intravenous injection set	Control clamp	3.2.5	4.4.7	S-3	*
1-Intravenous injection set	Freedom from pyrogen	3.5	4.4.8	USP	*
1-Intravenous injection set	Sterility	3.5	4.4.9	USP	*
1-Intravenous	Safety	3.5	4.4.10	USP	*

* Acceptance number shall be zero for all sample sizes.

**Each lot of product need not be biologically tested provided evidence is furnished that samples of the formulation and identical method of processing had previously been tested, that they comply with USP requirements for Class II plastics and that they are the same as being offered on this contract.

4.2.5 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination.

4.3.1 Examination of the intravenous injection set. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Examination, however, shall not be restricted to the classified possible defects listed in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	End protectors missing or dislodged.
102	Cotton missing (or inadequate amount) from open type protector.
103	Not free of bloom or tackiness.
104	Set incomplete.
105	Style and design not as specified.
106	Presence of foreign matter.
107	Set not free of holes or cuts.
Minor	
201	Instructions missing, incomplete or illegible.

4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

4.4 Tests.

4.4.1 Heavy metals extractable test. A 2 gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm., and eluted for 72 hours at 50°C., 24 hours at 70°C., or 60 minutes at 121°C. in one menstrua; 0.9% sodium chloride. Each gram of the material shall be covered with 5 millliliters (ml.) of the menstruum. The menstrua are collected, agitated after elution, and then submitted to spectographic atomic absorption spectrophotometric or specific wet analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg. from 200 g of test material dispersed in 1,000 ml. of elution menstruum.

4.4.1.1 Alternate heavy metals extractable test. Accurately weigh a 5-gram sample for each of the metals (barium, cadmium or lead) into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650°C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard solutions of 1, 2, and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and a 10X scale expansion on a spectrophotometer, such as Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculations:

A = Absorbance

$$F = \text{Factor} = \frac{(\text{Concentration standard in ppm Ba, Cd, or Pb})^*}{\text{absorbance}}$$

W = Sample weight in grams

*The three standards are averaged to give ppm/unit A.

4.4.2 Biological test. The plastics shall be tested in accordance with the USP for Class II plastics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

4.4.3 Drop test. The intravenous injection set shall be tested at room temperature, using water at a rate of 100 ± 5 drops per minute. The number of drops required to deliver 2 ml. of water shall be counted, and the drops per ml. calculated from this value.

4.4.4 Piercing device durability test. Samples of finished sets, after sterilization shall be used for this test. The piercing device (on assembled set) shall be inserted into, and removed from, a standard solution container (specified in 3.2.2) in the normal manner. This operation shall be repeated twice. The piercing device shall then be immersed in a solution of 5 percent dextrose in isotonic sodium chloride solution (in accordance with requirements of the USP monograph for "Dextrose and Sodium Chloride Injections") for 4 hours. The piercing device shall then be removed from the solution and immediately inserted into a standard solution container as specified in 3.2.2.

4.4.5 Leakage and separation test. The test shall be applied to those samples subjected to the piercing device durability test (see 4.4.4). The needle, and the cap containing the air inletting bacterial filter shall be removed. The plastic needle adapter of the set shall be fitted into a blind needle cavity. The set shall than be immersed in water and subjected to a pneumatic pressure of 8 pounds per square inch gage (psig) for 5 seconds. The pressure shall be applied through the piercing device and the set observed for separation and leakage.

4.4.5.1 Leakage and drip test. An infusion rate of 40 ± 5 drops per minute shall be established on the intravenous injection set when set-up as shown in Figure 2. The solution bag shall be squeezed by hand to 100 mm ± 10 of Hg maintained for 5 seconds, observed for leakage after which the infusion rate shall be measured which shall not vary more than 10 drops per minute from the original infusion rate setting.

4.4.6 Reseal test. The rubber diaphragm shall reseal without a falling drop within 10 seconds after puncture with a non-lubricated 18 gage needle when subjected to an internal pressure (hydrostatic pressure) of 4 p.s.i.g. for 30 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

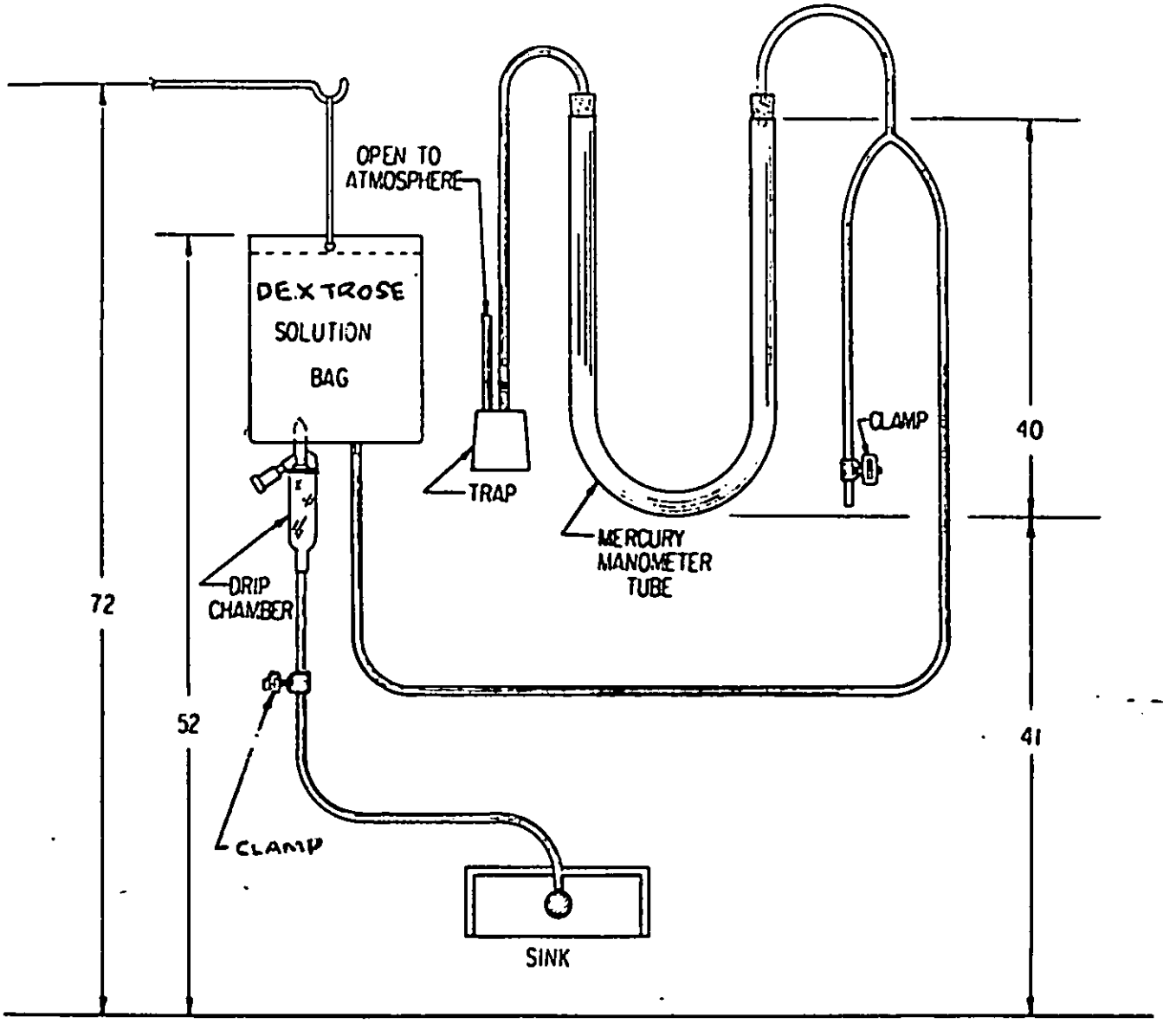


FIGURE 2
LEAKAGE AND DRIP RATE SET-UP

NOTE:
DIMENSIONS ARE IN INCHES AND ARE SUGGESTED ONLY.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

4.4.7 Clamping device performance test. The assembled set (except for needle) shall be connected, in the normal manner, to a standard solution container as specified in 3.2.2 except, as an alternate, the container may be filled with saline solution or water. The container, with set connected thereto, shall be suspended, in the normal manner, at a sufficient height to permit the entire length of the tubing of the set to hang vertically. The ability of the clamp to accomplish complete shut off, to regulate flow, and with a constant head of water to maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic-tubing within two inches above the molded rubber part and shall meet the requirement specified in 3.2.5. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the needle adapter.

4.4.8 Freedom from pyrogens. Samples of the finished sets shall be tested for freedom from pyrogens, in accordance with the USP, for transfusion and infusion assemblies.

4.4.9 Sterility test. The intravenous injection set shall be tested for sterility in accordance with the method specified in the USP for parenteral devices using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.9.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II, Sampling Criteria for Tests.

4.4.9.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55°-65°C
Dry Heat	B. stearothermophilus	55°-65°C
	B. subtilis var. niger (or globigii)	30°-35°C
Gas	B. subtilis var. niger (or globigii)	30°-35°C
Radiation	B. pumilus	30°-35°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.9.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.9 Sterility test and 4.4.9.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.9.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.9.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.9.4 Product testing. Testing for sterility of the product after receipt by the government shall be without biological indicator and in accordance with the USP.

4.4.10 Safety test. The finished sets shall be tested for safety in accordance with the USP, for transfusion and infusion assemblies.

4.4.11 Solution flow rate. The finished intravenous injection set shall be connected to a standard solution container as specified in 3.2.2 in the normal manner, the entire assembly inverted and suspended so that there is a 30 inch column of fluid in the set and container and no restriction of flow imposed by the needle or the clamp. The time necessary for passage of 100 cc. of fluid through the entire set shall be determined.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing forty-eight (48) sets, as specified, constitutes one unit of issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

5.1.2 Level A.

5.1.2.1 Individual box. Each set shall be packaged in a sealed box. At the option of the contractor, each box may have a die-cut window and be overwrapped in clear plastic or cellophane. Wrapper of box shall be sealed to prevent accidental opening. Wrapper shall have a tear strip for ease of opening.

5.1.2.2 Unit package. Forty eight individual boxes of one type only shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, class weather-resistant. The box shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of the bottom flaps with adhesive or metal stitches is permitted.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Four unit packages containing sets of one type only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class weather-resistant, grade V3c. Closure, waterproofing and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Four unit packages containing sets of one type only shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Individual box. In addition to the contractor's commercial markings, each box shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act.

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number, sterility lot (control) number and name of contractor shall shown. In addition, marking shall include the method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) which shall be shown following the level of packaging and date of packaging. Example: "A-2/75-G". Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot (control) number shall be shown. In addition, the method of sterilization as specified in 5.3.2 shall be shown.

6. NOTES

6.1 Intended use. The intravenous injection set is intended for accurate administration of single parenteral solution from a flexible plastic solution container.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Applicable NSN.

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

NSN 6515-00-115-0032	INTRAVENOUS INJECTION SET, with Flexible Drip Chamber, Without Needle, Disposable, 48s (Type I).
NSN 6515-00-140-0888	INTRAVENOUS INJECTION SET, with Flexible Drip Chamber, Without Needle, Disposable, 48s (Type II)

This specification does not include all types, sizes, or grades as indicated by the title or all types which are commercially available.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1975
is

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: *In a continuing effort to make our standardization documents better,* the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 6, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC-ATTS
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:13Z
Check the source to verify that this is the current version before use.